

OS LAB 08 & 09

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Task 1a (Code)

```
#include <stdio.h>
#include <pthread.h>

int shared = 0;

void* increment(void* arg) {
    for(int i = 0; i < 5; i++) {
        int temp = shared;
        temp++;
        shared = temp;
    }
    return NULL;
}

int main() {
    pthread_t t1, t2;

    pthread_create(&t1, NULL, increment, NULL);
    pthread_create(&t2, NULL, increment, NULL);

    pthread_join(t1, NULL);
    pthread_join(t2, NULL);

    printf("Final value (without sync): %d\n", shared);

    return 0;
}
```

Task 1b (Code)

```
#include <stdio.h>
#include <pthread.h>

int shared = 0;
pthread_mutex_t lock;

void* increment(void* arg) {
    for(int i = 0; i < 5; i++) {
        pthread_mutex_lock(&lock);

        shared++;

        pthread_mutex_unlock(&lock);
    }
}
```

```
    return NULL;
}

int main() {
    pthread_t t1, t2;

    pthread_mutex_init(&lock, NULL);

    pthread_create(&t1, NULL, increment, NULL);
    pthread_create(&t2, NULL, increment, NULL);

    pthread_join(t1, NULL);
    pthread_join(t2, NULL);

    printf("Final value (with mutex): %d\n", shared);

    pthread_mutex_destroy(&lock);

    return 0;
}
```

Task 1 (Screenshot)



```
system@system:~/Desktop$ gedit question_1a.c
system@system:~/Desktop$ gedit question_1b.c
system@system:~/Desktop$ gcc question_1a.c -o question_1a
system@system:~/Desktop$ gcc question_1b.c -o question_1b
system@system:~/Desktop$ ./question_1a
Final value (without sync): 10
system@system:~/Desktop$ ./question_1b
Final value (with mutex): 10
```

Task 2 (Code)

```
#include <stdio.h>
#include <signal.h>
#include <unistd.h>
#include <stdlib.h>

void handle_sigint(int sig) {
    printf("\nSIGINT received! Exiting program safely...\n");
    exit(0);
}

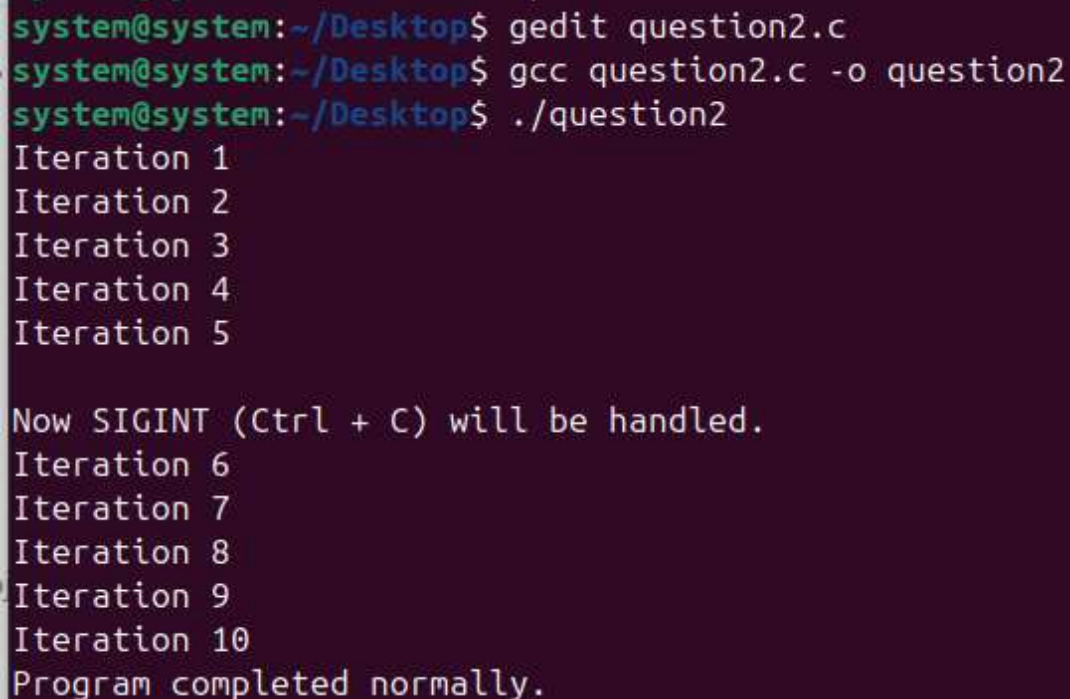
int main() {
    signal(SIGINT, SIG_IGN);

    for(int i = 1; i <= 10; i++) {
        printf("Iteration %d\n", i);
        sleep(1);

        if(i == 5) {
            printf("\nNow SIGINT (Ctrl + C) will be handled.\n");
            signal(SIGINT, handle_sigint);
        }
    }

    printf("Program completed normally.\n");
    return 0;
}
```

Task 2 (Screenshot)



```
system@system:~/Desktop$ gedit question2.c
system@system:~/Desktop$ gcc question2.c -o question2
system@system:~/Desktop$ ./question2
Iteration 1
Iteration 2
Iteration 3
Iteration 4
Iteration 5

Now SIGINT (Ctrl + C) will be handled.
Iteration 6
Iteration 7
Iteration 8
Iteration 9
Iteration 10
Program completed normally.
```

Task 3 (Code)

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <signal.h>
#include <sys/wait.h>

void sigchld_handler(int sig) {
    int status;
    pid_t pid;

    while ((pid = waitpid(-1, &status, WNOHANG)) > 0) {
        printf("Child process with PID %d terminated\n", pid);
    }
}

int main() {
    signal(SIGCHLD, sigchld_handler);

    for (int i = 0; i < 3; i++) {
        pid_t pid = fork();

        if (pid == 0) {
            printf("Child %d started with PID %d\n", i + 1, getpid());
            sleep(3 + i * 2);
            printf("Child %d (PID %d) exiting\n", i + 1, getpid());
            exit(0);
        }
    }

    while (1) {
        printf("Server running and monitoring workers...\n");
        sleep(2);
    }

    return 0;
}
```

Task 3 (Screenshots)

```
system@system:~/Desktop$ gedit question3.c
system@system:~/Desktop$ gcc question3.c -o question3
system@system:~/Desktop$ ./question3
```

```
Child 3 started with PID 6004
Child 1 started with PID 6002
Server running and monitoring workers...
Child 1 (PID 6002) exiting
Child process with PID 6002 terminated
Server running and monitoring workers...
Child 2 (PID 6003) exiting
Child process with PID 6003 terminated
Server running and monitoring workers...
Child 3 (PID 6004) exiting
Server running and monitoring workers...
Server running and monitoring workers...
Child process with PID 6004 terminated
Server running and monitoring workers...
Server running and monitoring workers...
```

```
system@system:~$ cd Desktop
system@system:~/Desktop$ kill -9 6003
bash: kill: (6003) - No such process
system@system:~/Desktop$
```